

$$\begin{bmatrix} \cos 90^\circ & \sin 90^\circ \\ -\sin 90^\circ & \cos 90^\circ \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

<https://xkcd.com/184/>

Classic Mechs EP3: Eigenvector and Higher Order Linear ODE Systems

Classic Mechs EP3.1: higher order ODE systems

Learning objectives

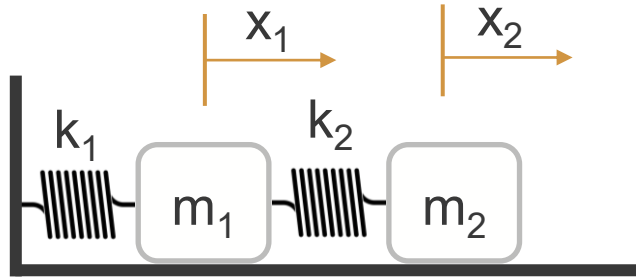
- 1) Work on double harmonic oscillators and derive the slightly messier 4th order ODE.
- 2) Back to the 1st order land: derive the general state space matrix form for higher order linear ODE systems.
- 3) Motivation to use the eigenvalue and eigenvector method to solve such systems of equations.

What's next for next week?

Let's start to move into the Lagrange land, and revisit our old friend of rotational motion, energy, conservative force... along the way.

Double harmonic oscillators = 4th order ODE

Double harmonic oscillators case A:



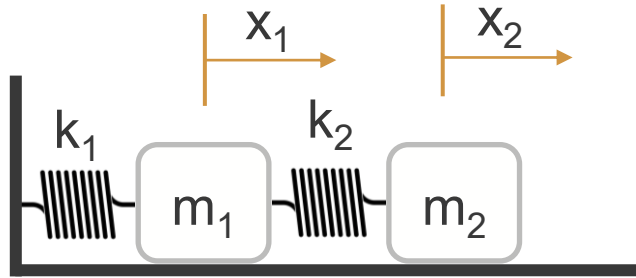
$$m_1 \ddot{x}_1 = -k_1 x_1 + k_2 (x_2 - x_1)$$

$$m_2 \ddot{x}_2 = -k_2 (x_2 - x_1)$$

→ Solve for $x_1 = f(x_2, \ddot{x}_2)$ } $\ddot{x}_1 = g(x_2, \ddot{x}_2, x_2^{(4)})$

Double harmonic oscillators = 4th order ODE

Double harmonic oscillators case A:



$$m_1 \ddot{x}_1 = -k_1 x_1 + k_2 (x_2 - x_1)$$

$$m_2 \ddot{x}_2 = -k_2 (x_2 - x_1)$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{v}_1 \\ \dot{x}_2 \\ \dot{v}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{k_1+k_2}{m_1} & 0 & \frac{k_2}{m_1} & 0 \\ 0 & 0 & 0 & 1 \\ \frac{k_2}{m_2} & 0 & -\frac{k_2}{m_2} & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ v_1 \\ x_2 \\ v_2 \end{bmatrix}$$

1st order system of equations not so bad now, heh?

Use systems of equations to tackle higher order systems

$a_n x^{(n)} + a_{n-1} x^{(n-1)} + \dots + a_2 \ddot{x} + a_1 \dot{x} + a_0 x = 0$ General form of higher order linear ODE
(guess $e^{\lambda t}$ again...)

$a_n \lambda^n + a_{n-1} \lambda^{n-1} + \dots + a_2 \lambda^2 + a_1 \lambda + a_0 = 0$ Characteristic equation

$X(t) = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t} + \dots + C_n e^{\lambda_n t}$ General form of solution

How many initial conditions we need to determine the constants of $C_1, C_2, \dots, C_n$?

How to write the higher order ODE in matrix system from?

$$a_n x^{(n)} + a_{n-1} x^{(n-1)} + \dots + a_2 \ddot{x} + a_1 \dot{x} + a_0 x = 0$$

Introducing:

$$x_1 = x$$

$$x_2 = \dot{x}$$

$$x_3 = \ddot{x}$$

$$\vdots$$

$$x_n = x^{(n-1)}$$

That's

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = x_4$$

$$\vdots$$

$$\dot{x}_n = x^{(n)} = -(a_{n-1} x^{(n-1)} + \dots + a_2 \ddot{x} + a_1 \dot{x} + a_0 x)$$

$$\dot{\underline{x}} = \underline{A} \underline{x}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ 0 & 0 & 0 & \ddots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 0 & 1 \\ -a_0 & -a_1 & -a_2 & \dots & -a_{n-2} & -a_{n-1} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix}$$



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This (a 4th order to state space matrix) will show up in mid-term.

It's time to solve general systems of equations: $\underline{\dot{x}} = \underline{D} \underline{x}$

Case 1: uncoupled/decoupled systems, really really easy!

Just like: $\dot{x}_n = \lambda_n x_n \rightarrow x_n(t) = e^{\lambda_n t} x_n(0)$

We will get:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & 0 \\ & \vdots & \ddots & \vdots \\ 0 & 0 & & 0 \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} \rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} (t) = \begin{bmatrix} e^{\lambda_1 t} & 0 & \cdots & 0 \\ 0 & e^{\lambda_2 t} & & 0 \\ & \vdots & \ddots & \vdots \\ 0 & 0 & & 0 \\ 0 & 0 & \cdots & e^{\lambda_n t} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} (0)$$

Why they are eigenvalues already?



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Case 1: uncoupled/decoupled systems is really easy: $\underline{\dot{x}} = \underline{D} \underline{x}$

Matrix exponential?

Case 1: uncoupled/decoupled systems

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & 0 \\ & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \\ 0 & 0 & & \lambda_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} \rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} (t) = \overbrace{\begin{bmatrix} e^{\lambda_1 t} & 0 & \cdots & 0 \\ 0 & e^{\lambda_2 t} & & 0 \\ & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \\ 0 & 0 & & e^{\lambda_n t} \end{bmatrix}}^{e^{\underline{D}t}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} (0)}_{\text{Initial condition}}$$

$$\dot{x}_n = \lambda_n x_n \rightarrow x_n(t) = e^{\lambda_n t} \underbrace{x_n(0)}_{\text{Initial condition}}$$



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Matrix exponential?

https://en.wikipedia.org/wiki/Matrix_exponential

Diagonalizable case [\[edit\]](#)

If a matrix is [diagonal](#):

$$A = \begin{bmatrix} a_1 & 0 & \cdots & 0 \\ 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_n \end{bmatrix},$$

then its exponential can be obtained by exponentiating each entry on the main diagonal:

$$e^A = \begin{bmatrix} e^{a_1} & 0 & \cdots & 0 \\ 0 & e^{a_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e^{a_n} \end{bmatrix}.$$

This result also allows one to exponentiate [diagonalizable matrices](#). If

$$A = UDU^{-1}$$

then

$$e^A = Ue^DU^{-1},$$

which is especially easy to compute when D is diagonal.

It's time to solve general systems of equations:

Case 2: for general system $\dot{\underline{x}} = \underline{A} \underline{x}$

Introduce a change of coordinates: $\underline{x} = \underline{T} \underline{z}$

That will diagonalizes the ODE: $\dot{\underline{z}} = \underline{D} \underline{z} \rightarrow \underline{z}(t) = e^{\underline{D}t} \underline{z}(0)$

Chain rule, and
T not time dependent

$$\begin{aligned} \underline{x} = \underline{T} \underline{z} &\rightarrow \dot{\underline{x}} = \cancel{\dot{\underline{T}}} \underline{z} + \underline{T} \dot{\underline{z}} = \underline{T} \dot{\underline{z}} \rightarrow \dot{\underline{x}} = \underline{T} \dot{\underline{z}} \\ \dot{\underline{x}} &= \underline{A} \underline{x} \end{aligned} \left. \vphantom{\begin{aligned} \underline{x} = \underline{T} \underline{z} \\ \dot{\underline{x}} = \underline{A} \underline{x} \end{aligned}} \right\} \rightarrow \begin{aligned} \underline{T} \dot{\underline{z}} &= \underline{A} \underline{x} \\ \underline{x} &= \underline{T} \underline{z} \end{aligned} \left. \vphantom{\begin{aligned} \underline{T} \dot{\underline{z}} &= \underline{A} \underline{x} \\ \underline{x} &= \underline{T} \underline{z} \end{aligned}} \right\} \rightarrow \underline{T} \dot{\underline{z}} = \underline{A} \underline{T} \underline{z} \left. \vphantom{\underline{T} \dot{\underline{z}} = \underline{A} \underline{T} \underline{z}} \right\} \rightarrow \dot{\underline{z}} = \underline{T}^{-1} \underline{A} \underline{T} \underline{z}$$

https://en.wikipedia.org/wiki/Invertible_matrix



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Case 2: turn general systems into diagonal form so it's easy too

Case 2: for general system $\underline{\dot{x}} = \underline{A} \underline{x}$

Introduce a change of coordinates: $\underline{x} = \underline{T} \underline{z}$

That will diagonalizes the ODE: $\underline{\dot{z}} = \underline{D} \underline{z} \rightarrow \underline{z}(t) = e^{\underline{D}t} \underline{z}(0)$

$$\left. \begin{array}{l} \underline{\dot{z}} = \underline{T}^{-1} \underline{A} \underline{T} \underline{z} \\ \underline{\dot{z}} = \underline{D} \underline{z} \end{array} \right\} \rightarrow \underline{T}^{-1} \underline{A} \underline{T} = \underline{D}$$

https://en.wikipedia.org/wiki/Invertible_matrix

$$\left. \begin{array}{l} \underline{T}^{-1} \underline{A} \underline{T} = \underline{D} \\ \underline{A} \underline{T} = \underline{T} \underline{D} \end{array} \right\}$$

Eigenvalues already!
T is made with columns of Eigenvectors

Finally, Eigenvalues and Eigenvectors

$$\underline{\underline{A}} \underline{\underline{T}} = \underline{\underline{T}} \underline{\underline{D}}$$

Eigenvalues already!

T is made with columns of Eigenvectors

$$\underline{\underline{A}} \begin{bmatrix} \underline{t}_1 & \underline{t}_2 & \dots & \underline{t}_n \end{bmatrix} = \begin{bmatrix} \underline{t}_1 & \underline{t}_2 & \dots & \underline{t}_n \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & & 0 \\ & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \\ 0 & 0 & & \lambda_n \end{bmatrix}$$

$$= \begin{bmatrix} \lambda_1 \underline{t}_1 & \lambda_2 \underline{t}_2 & \dots & \lambda_n \underline{t}_n \end{bmatrix}$$

$$\underline{\underline{A}} \underline{t}_n = \lambda_n \underline{t}_n$$

For specific eigenvalue λ_n , there's a corresponding eigenvector $\underline{t}_n$!



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Example of 2x2 systems

$$\dot{\underline{x}} = \underline{A} \underline{x} \quad \underline{A} = \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix} \quad \underline{T} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad \underline{D} = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$$